Jeffrey Buskirk | United States Citizen
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EDUCATION

University of Arizona | College of Engineering | Tucson, AZ
Aerospace Engineering | GPA: 3.91 | Expected Graduation: May 2028 | W.A. Franke Honors College

SKILLS

Simulation & Analysis: CFD (ANSYS Fluent, CFX, ADS, CFD++, STAR-CCM+), meshing (Pointwise, HELDEN), FEA ANSYS Static Structural, Abaqus), Thermal (ANSYS Transient Thermal, NASA CEA)
CAD/CAM & Electrical: NX, SolidWorks, Fusion 360, Onshape and KiCad
Programming & Hardware: MATLAB, Python, LabVIEW; Linux/HPC workflows; data automation & post-processing tooling; machining (lathe, mill), additive manufacturing, I²C sensor integration
AWARDS: 1st Place Category A – FAR-OUT Competition (Friends of Amateur Rocketry, Oxidizer Uninhibited Tournament), 2× Honeywell Aerospace Additive Manufacturing Contest Winner, Innovation Award (VEX Robotics Worlds), Forklift Certified

PROFESSIONAL EXPERIENCE

Turbomachinery Engineering Intern – Boom Supersonic | Summer 2026

- Owned a mission-critical low-pressure compressor (LPC) stall-margin issue; traced the root cause to high Rotor 1 incidence and ran a design-of-experiments study on IGV and rotor leading-edge angles across 60–100% corrected speed, recovering positive stall margin at off-design conditions without sacrificing 100% speed efficiency.
- Built an automated CFD workflow and AI-integrated dashboard to triage DOE cases by performance impact and rapidly run unsteady CFD on top candidates, including automated speed-line point submission.
- Redesigned the aft-turbine outlet diffuser for high-temperature, high-Mach flow, meeting the <1.77% pressure-loss target while cutting cost ~50% by switching from complicated geometry that would require spin casting to sheet metal.

Manufacturing and Test Engineering Intern – R3 Aerospace | Winter 2026 – Ongoing

- Designed and validated high-pressure solid rocket motor bulkheads using ANSYS FEA through hydro-proof and static-fire testing; developed test infrastructure supporting medium-scale SRMs for defense customers.
- Bridged design and production, working directly with manufacturing to resolve fabrication and tolerance challenges leading to successful static-fire and flight testing.

Chief Engineer, Former Propulsion Manager – Wildcat Rocket Engineering Club | Fall 2025 – Ongoing

- Led the first team to successfully launch a liquid rocket twice in the FAR-OUT competition, earning 1st place in Category A.
- Own end-to-end development of a 1,400 lbf nitrous–ethanol liquid engine and a student-designed N_2O/E98 rocket (9.5:1 TWR at 600 lbf max thrust) — design, manufacturing, a custom test stand, and cold-flow/static-fire testing.
- Spearheaded qualification testing for competition eligibility — hydrostatic proofs to 1500 psi and successful cold-flow tests characterizing electronic-dump and injector behavior.
- Led the failure investigation after a hard start on the 1,400 lbf engine's first hot-fire; used ANSYS to identify the combustion-instability driver and redesigned the injector, enabling successful subsequent testing.
- Conducted CFD/FEA nozzle and injector trade studies with NASA CEA, ANSYS Fluent and Transient Thermal, quantifying wall heat flux to size wall thickness and selecting nozzle material scheme to optimize cost, machinability, and reusability.
- Trained the team in ANSYS, CAD, and machining; secured ANSYS, SendCutSend, and Bolt line (A Stoke Company) sponsorships, and ran program logistics, bill-of-materials tracking, and engineering–administration collaboration in Bolt line.

Engineering Research Associate – The Space Analog for the Moon and Mars | Fall 2025 – Summer 2026

- Led a 3-person team delivering a Concept Space Station CO_2-removal prototype for Paragon Space Development and ICES; designed and built the AC electrical control system (manual override + relay automation) for adsorption/desorption cycling, producing system-effectiveness and data. Led Mechanical design effort ensuring the chamber could cycle between extreme temperature and vacuum conditions.

Turbomachinery Engineering Intern – Honeywell Aerospace (Fans and Compressors) | Summer 2025

- Analyzed compressor performance maps in ANSYS CFX (post-processed in MATLAB/Excel), improving adiabatic efficiency ~0.5% via clearance and bleed-air studies for Honeywell's next generation turbofan engine.
- Built a Thermodynamic Performance map-to-CFX benchmarking capability new to the team for surface-roughness studies (~1% efficiency gain toward the 10,000 lbf thrust target); ran large-scale CFD on Linux HPC with MATLAB tools that cut post-processing runtime 20%.

Test Engineering Intern – Honeywell Aerospace | Summer 2024

- Designed test articles for the updated RS-25 main fuel/oxidizer valves (NASA Artemis SLS) in NX with cryogenic test protocols and ran actuator testing for NASA Gateway docking hardware in cleanroom thermal-cycle environments.
- Programmed (C++), machined, and designed controllers for EVTOL actuation qualification rigs — presented to management and adopted by the customer; also supported TVC actuator testing on a DoD contract.

PUBLICATIONS

Hentzen, G., Staats, K., Ackerman, A., Buskirk, J., Phoebe, B., & Knox, J. C. "Further Design and Initial AI&T of the Carbon Dioxide Removal System for SAM." 55th International Conference on Environmental Systems (ICES-2026-524), 2026.